

```
praveenl [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
18 19 20 21 22 23 24 22 23 24 25 26 27 28 20 21 22 23 24 25 26
25 26 27 28 29 30 31 29 30 27 28 29 30 31

[praveenl213@localhost ~]$ cal 01 2026
    January 2026
Su Mo Tu We Th Fr Sa
                1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

[praveenl213@localhost ~]$ echo os
os
[praveenl213@localhost ~]$ echo ""

[praveenl213@localhost ~]$ echo "Hello everyone"
Hello everyone
[praveenl213@localhost ~]$ bc -l
bc 1.07.1
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This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
100-2
98
200/2
100.000000000000000000000000
quit
[praveenl213@localhost ~]$ who
praveenl213 tty2      2026-01-23 11:07 (local)
[praveenl213@localhost ~]$ whoami
praveenl213
[praveenl213@localhost ~]$ id
uid=1000(praveenl213) gid=1000(praveenl213) groups=1000(praveenl213) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[praveenl213@localhost ~]$ man ls
[praveenl213@localhost ~]$ manm cd
bash: manm: command not found...
Similar command is: 'man'
[praveenl213@localhost ~]$ man cd
[praveenl213@localhost ~]$ ps
  PID TTY          TIME CMD
 2804 pts/0    00:00:00 bash
 2984 pts/0    00:00:00 ps
[praveenl213@localhost ~]$ uname -a
Linux localhost.localdomain 5.14.0-661.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Jan 16 09:19:22 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
[praveenl213@localhost ~]$
```

```
File Machine View Input Devices Help
[praveenl213@localhost ~]$ ls
a Desktop Documents Downloads Music Pictures Public Templates Videos
[praveenl213@localhost ~]$ mkdir os.txt
[praveenl213@localhost ~]$ ls
a Documents Music Pictures Templates
Desktop Downloads os.txt Public Videos
[praveenl213@localhost ~]$ cd a/
[praveenl213@localhost a]$ ls
a b c os os.txt
[praveenl213@localhost a]$ mkdir a b c
mkdir: cannot create directory 'a': File exists
mkdir: cannot create directory 'b': File exists
mkdir: cannot create directory 'c': File exists
[praveenl213@localhost a]$ ls
a b c os os.txt
[praveenl213@localhost a]$ mkdir -p x/y/z
[praveenl213@localhost a]$ cd x/
bash: cd x/: No such file or directory
[praveenl213@localhost a]$ cd x/
[praveenl213@localhost x]$ cd y/
[praveenl213@localhost y]$ cd z/
[praveenl213@localhost z]$ cd ../
[praveenl213@localhost y]$ cd ..
[praveenl213@localhost x]$ cd ../
[praveenl213@localhost a]$ cd x/
[praveenl213@localhost x]$ cd y/
[praveenl213@localhost y]$ rmdir z
[praveenl213@localhost y]$ ls
[praveenl213@localhost y]$ cd ../
[praveenl213@localhost x]$ rmdir y
[praveenl213@localhost x]$ ls
[praveenl213@localhost x]$ cd ../
[praveenl213@localhost a]$ rmdir x
[praveenl213@localhost a]$ ls
a b c os os.txt
[praveenl213@localhost a]$ mkdir -p x/y/z
[praveenl213@localhost a]$ rmdir _rf x
rmdir: failed to remove '_rf': No such file or directory
rmdir: failed to remove 'x': Directory not empty
[praveenl213@localhost a]$ -rf x
bash: -rf: command not found...
[praveenl213@localhost a]$ rm -rf x
[praveenl213@localhost a]$ ls
a b c os os.txt
[praveenl213@localhost a]$ cat > os
^Z
[1]+  Stopped                  cat > os
```

NAME

bash, :, ., [, alias, bg, bind, break, builtin, caller, cd, command, compgen, complete, compopt, continue, declare, dirs, disown, echo, enable, eval, exec, exit, export, false, fc, fg, getopts, hash, help, history, jobs, kill, let, local, logout, mapfile, popd, printf, pushd, pwd, read, readonly, return, set, shift, shopt, source, suspend, test, times, trap, true, type, typeset, ulimit, umask, unalias, unset, wait - bash built-in commands, see **bash**(1)

BASH BUILTIN COMMANDS

Unless otherwise noted, each builtin command documented in this section as accepting options preceded by - accepts -- to signify the end of the options. The :, **true**, **false**, and **test**/[builtins do not accept options and do not treat -- specially. The **exit**, **logout**, **return**, **break**, **continue**, **let**, and **shift** builtins accept and process arguments beginning with - without requiring --. Other builtins that accept arguments but are not specified as accepting options interpret arguments beginning with - as invalid options and require -- to prevent this interpretation.

: [arguments]

No effect; the command does nothing beyond expanding arguments and performing any specified redirections. The return status is zero.

. filename [arguments]

source filename [arguments]

Read and execute commands from filename in the current shell environment and return the exit status of the last command executed from filename. If filename does not contain a slash, filenames in **PATH** are used to find the directory containing filename. The file searched for in **PATH** need not be executable. When **bash** is not in posix mode, the current directory is searched if no file is found in **PATH**. If the **sourcepath** option to the **shopt** builtin command is turned off, the **PATH** is not searched. If any arguments are supplied, they become the positional parameters when filename is executed. Otherwise the positional parameters are unchanged. If the -T option is enabled, **source** inherits any trap on **DEBUG**; if it is not, any **DEBUG** trap string is saved and restored around the call to **source**, and **source** unsets the **DEBUG** trap while it executes. If -T is not set, and the sourced file changes the **DEBUG** trap, the new value is retained when **source** completes. The return status is the status of the last command exited within the script (0 if no commands are executed), and false if filename is not found or cannot be read.

alias [-p] [name[=value] ...]

Alias with no arguments or with the -p option prints the list of aliases in the form **alias name=value** on standard output. When arguments are supplied, an alias is defined for each name whose value is given. A trailing space in value causes the next word to be checked for alias substitution when the alias is expanded. For each name in the argument list for which no value is supplied, the name and value of the alias is printed. **Alias** returns true unless a name is given for which no alias has been defined.

bg [jobspec ...]

Resume each suspended job jobspec in the background, as if it had been started with &. If jobspec is not present, the shell's notion of the current job is used. **bg** jobspec returns 0 unless run when job control is disabled or, when run with job control enabled, any specified job-spec was not found or was started without job control.

bind [-m keymap] [-lpsvPSVX]

bind [-m keymap] [-q function] [-u function] [-r keyseq]

bind [-m keymap] -f filename

bind [-m keymap] -x keyseq:shell-command

bind [-m keymap] keyseq:function-name

bind [-m keymap] keyseq:readline-command

```
praveenl [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
[praveenl213@localhost ~]$ ls
a c Documents Music Pictures Templates
b Desktop Downloads os.txt Public Videos
[praveenl213@localhost ~]$ cal 2026
2026

January February March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 3 1 2 3 4 5 6 7 1 2 3 4 5 6 7
4 5 6 7 8 9 10 8 9 10 11 12 13 14 8 9 10 11 12 13 14
11 12 13 14 15 16 17 15 16 17 18 19 20 21 15 16 17 18 19 20 21
18 19 20 21 22 23 24 22 23 24 25 26 27 28 22 23 24 25 26 27 28
25 26 27 28 29 30 31 29 30 31

April May June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 3 4 1 2 1 2 3 4 5 6
5 6 7 8 9 10 11 3 4 5 6 7 8 9 7 8 9 10 11 12 13
12 13 14 15 16 17 18 10 11 12 13 14 15 16 14 15 16 17 18 19 20
19 20 21 22 23 24 25 17 18 19 20 21 22 23 21 22 23 24 25 26 27
26 27 28 29 30 24 25 26 27 28 29 30 28 29 30
31

July August September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 3 4 1 1 2 3 4 5
5 6 7 8 9 10 11 2 3 4 5 6 7 8 6 7 8 9 10 11 12
12 13 14 15 16 17 18 9 10 11 12 13 14 15 13 14 15 16 17 18 19
19 20 21 22 23 24 25 16 17 18 19 20 21 22 20 21 22 23 24 25 26
26 27 28 29 30 31 23 24 25 26 27 28 29 27 28 29 30
30 31

October November December
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 3 1 2 3 4 5 6 7 1 2 3 4 5
4 5 6 7 8 9 10 8 9 10 11 12 13 14 6 7 8 9 10 11 12
11 12 13 14 15 16 17 15 16 17 18 19 20 21 13 14 15 16 17 18 19
18 19 20 21 22 23 24 22 23 24 25 26 27 28 20 21 22 23 24 25 26
25 26 27 28 29 30 31 29 30 27 28 29 30 31

[praveenl213@localhost ~]$ cal 01 2026
January 2026
Su Mo Tu We Th Fr Sa
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```


File Machine View Input Devices Help

time and sort by name; otherwise: sort by access time, newest first

-U do not sort; list entries in directory order

-v natural sort of (version) numbers within text

-w, --width=COLS
set output width to COLS. 0 means no limit

-x list entries by lines instead of by columns

-X sort alphabetically by entry extension

-Z, --context
print any security context of each file

-1 list one file per line. Avoid '\n' with **-q** or **-b**

--help display this help and exit

--version
output version information and exit

The **SIZE** argument is an integer and optional unit (example: 10K is 10×1024). Units are K,M,G,T,P,E,Z,Y (powers of 1024) or KB,MB,... (powers of 1000). Binary prefixes can be used, too: KiB=K, MiB=M, and so on.

The **TIME_STYLE** argument can be full-iso, long-iso, iso, locale, or +FORMAT. **FORMAT** is interpreted like in date(1). If **FORMAT** is **FORMAT1<newline>FORMAT2**, then **FORMAT1** applies to non-recent files and **FORMAT2** to recent files. **TIME_STYLE** prefixed with 'posix-' takes effect only outside the POSIX locale. Also the **TIME_STYLE** environment variable sets the default style to use.

Using **color** to distinguish file types is disabled both by default and with **--color=never**. With **--color=auto**, **ls** emits color codes only when standard output is connected to a terminal. The **LS_COLORS** environment variable can change the settings. Use the **dircolors** command to set it.

Exit status:

0 if OK,

1 if minor problems (e.g., cannot access subdirectory),

Manual page **ls(1)** line 180/245 91% (press h for help or q to quit)

praveenl [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
--show-control-chars
    show nongraphic characters as-is (the default, unless program is
    'ls' and output is a terminal)

-Q, --quote-name
    enclose entry names in double quotes

--quoting-style=WORD
    use quoting style WORD for entry names: literal, locale, shell,
    shell-always, shell-escape, shell-escape-always, c, escape
    (overrides QUOTING_STYLE environment variable)

-r, --reverse
    reverse order while sorting

-R, --recursive
    list subdirectories recursively

-s, --size
    print the allocated size of each file, in blocks

-S
    sort by file size, largest first

--sort=WORD
    sort by WORD instead of name: none (-U), size (-S), time (-t),
    version (-v), extension (-X)

--time=WORD
    change the default of using modification times; access time
    (-u): atime, access, use; change time (-c): ctime, status; birth
    time: birth, creation;

    with -l, WORD determines which time to show; with --sort=time,
    sort by WORD (newest first)

--time-style=TIME_STYLE
    time/date format with -l; see TIME_STYLE below

-t
    sort by time, newest first; see --time

-T, --tabsize=COLS
    assume tab stops at each COLS instead of 8

-u
    with -lt: sort by, and show, access time; with -l: show access
```

Manual page ls(1) line 135/245 71% (press h for help or q to quit)

```
praveenl [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

--version
    output version information and exit

The SIZE argument is an integer and optional unit (example: 10K is
10*1024). Units are K,M,G,T,P,E,Z,Y (powers of 1024) or KB,MB,...
(powers of 1000). Binary prefixes can be used, too: KiB=K, MiB=M, and
so on.

The TIME_STYLE argument can be full-iso, long-iso, iso, locale, or
+FORMAT. FORMAT is interpreted like in date(1). If FORMAT is FOR-
MAT1<newline>FORMAT2, then FORMAT1 applies to non-recent files and FOR-
MAT2 to recent files. TIME_STYLE prefixed with 'posix-' takes effect
only outside the POSIX locale. Also the TIME_STYLE environment vari-
able sets the default style to use.

Using color to distinguish file types is disabled both by default and
with --color=never. With --color=auto, ls emits color codes only when
standard output is connected to a terminal. The LS_COLORS environment
variable can change the settings. Use the dircolors command to set it.

Exit status:
0      if OK,

1      if minor problems (e.g., cannot access subdirectory),

2      if serious trouble (e.g., cannot access command-line argument).

AUTHOR
    Written by Richard M. Stallman and David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

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    There is NO WARRANTY, to the extent permitted by law.

SEE ALSO
    Full documentation <https://www.gnu.org/software/coreutils/ls>
    or available locally via: info '(coreutils) ls invocation'

GNU coreutils 8.32      December 2024      LS(1)
Manual page ls(1) line 201/245 (END) (press h for help or q to quit)
```

praveenl [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
-d, --directory
    list directories themselves, not their contents

-D, --dired
    generate output designed for Emacs' dired mode

-f
    do not sort, enable -aU, disable -ls --color

-F, --classify
    append indicator (one of */=>@|) to entries

--file-type
    likewise, except do not append '*'

--format=WORD
    across -x, commas -m, horizontal -x, long -l, single-column -l,
    verbose -l, vertical -C

--full-time
    like -l --time-style=full-iso

-g
    like -l, but do not list owner

--group-directories-first
    group directories before files;

    can be augmented with a --sort option, but any use of
    --sort=none (-U) disables grouping

-G, --no-group
    in a long listing, don't print group names

-h, --human-readable
    with -l and -s, print sizes like 1K 234M 2G etc.

--si
    likewise, but use powers of 1000 not 1024

-H, --dereference-command-line
    follow symbolic links listed on the command line

--dereference-command-line-symlink-to-dir
    follow each command line symbolic link

    that points to a directory
```

Manual page ls(1) line 45/245 34% (press h for help or q to quit)


```
praveenl [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
LS(1) User Commands LS(1)

NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current directory by default).
    Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        do not ignore entries starting with .

    -A, --almost-all
        do not list implied . and ..

    --author
        with -l, print the author of each file

    -b, --escape
        print C-style escapes for nongraphic characters

    --block-size=SIZE
        with -l, scale sizes by SIZE when printing them; e.g.,
        '--block-size=M'; see SIZE format below

    -B, --ignore-backups
        do not list implied entries ending with ~

    -c
        with -lt: sort by, and show, ctime (time of last modification of
        file status information); with -l: show ctime and sort by name;
        otherwise: sort by ctime, newest first

    -C
        list entries by columns

    --color[=WHEN]
        colorize the output; WHEN can be 'always' (default if omitted),
        'auto', or 'never'; more info below
```

praveenl [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
--hide=PATTERN
do not list implied entries matching shell PATTERN (overridden
by -a or -A)

--hyperlink[=WHEN]
hyperlink file names; WHEN can be 'always' (default if omitted),
'auto', or 'never'

--indicator-style=WORD
append indicator with style WORD to entry names: none (default),
slash (-p), file-type (--file-type), classify (-F)

-i, --inode
print the index number of each file

-I, --ignore=PATTERN
do not list implied entries matching shell PATTERN

-k, --kibibytes
default to 1024-byte blocks for disk usage; used only with -s
and per directory totals

-l use a long listing format

-L, --dereference
when showing file information for a symbolic link, show informa-
tion for the file the link references rather than for the link
itself

-m fill width with a comma separated list of entries

-n, --numeric-uid-gid
like -l, but list numeric user and group IDs

-N, --literal
print entry names without quoting

-o like -l, but do not list group information

-p, --indicator-style=slash
append / indicator to directories

-q, --hide-control-chars
print ? instead of nongraphic characters
```

Manual page ls(1) line 90/245 52% (press h for help or q to quit)

```
File Machine View Input Devices Help
[praveenl213@localhost ~]$ ls
a Desktop Documents Downloads Music Pictures Public Templates Videos
[praveenl213@localhost ~]$ mkdir os.txt
[praveenl213@localhost ~]$ ls
a Documents Music Pictures Templates
Desktop Downloads os.txt Public Videos
[praveenl213@localhost ~]$ cd a/
[praveenl213@localhost a]$ ls
a b c os os.txt
[praveenl213@localhost a]$ mkdir a b c
mkdir: cannot create directory 'a': File exists
mkdir: cannot create directory 'b': File exists
mkdir: cannot create directory 'c': File exists
[praveenl213@localhost a]$ ls
a b c os os.txt
[praveenl213@localhost a]$ mkdir -p x/y/z
[praveenl213@localhost a]$ cd x/
bash: cd x/: No such file or directory
[praveenl213@localhost a]$ cd x/
[praveenl213@localhost x]$ cd y/
[praveenl213@localhost y]$ cd z/
[praveenl213@localhost z]$ cd ../
[praveenl213@localhost y]$ cd ..
[praveenl213@localhost x]$ cd ../
[praveenl213@localhost a]$ cd x/
[praveenl213@localhost x]$ cd y/
[praveenl213@localhost y]$ rmdir z
[praveenl213@localhost y]$ ls
[praveenl213@localhost y]$ cd ../
[praveenl213@localhost x]$ rmdir y
[praveenl213@localhost x]$ ls
[praveenl213@localhost x]$ cd ../
[praveenl213@localhost a]$ rmdir x
[praveenl213@localhost a]$ ls
a b c os os.txt
[praveenl213@localhost a]$ mkdir -p x/y/z
[praveenl213@localhost a]$ rmdir _rf x
rmdir: failed to remove '_rf': No such file or directory
rmdir: failed to remove 'x': Directory not empty
[praveenl213@localhost a]$ -rf x
bash: -rf: command not found...
[praveenl213@localhost a]$ rm -rf x
[praveenl213@localhost a]$ ls
a b c os os.txt
[praveenl213@localhost a]$ cat > os
^Z
[1]+  Stopped                  cat > os
```